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Political and Eugenic Implications of Dating App Algorithms

A warm, dark bar filled with young adults chatting and dancing. Men and women scan the crowds, looking for someone to strike up a conversation with. They take in the whole room, noticing who smiles back and looks approachable, and single in on someone worth taking a chance on. Online, that scene looks very different. Cute? Swipe right, send a message. Not so cute? Swipe left, never to be seen again. In less than ten minutes, a roster of compatible partners, at least those deemed so by a hidden algorithm, sits in your inbox. Dating apps present users with photos, filters, and irreversible swipes as a window to countless eligible partners who have been curated by a model behind the scenes. The presence of someone on your dating feed is determined by automated decision making (ADM) systems, who use behavioral and demographic data to filter and rank profiles. Preemptive computing algorithms take this even further and predict what users will find desirable, then adjust the deck accordingly before any explicit user choice is made. As more romantic relationships are formed via dating apps than any other organic medium, including 39% of heterosexual couples in 2017, dating apps have become a sociotechnical system that forms structural patterns of intimacy beyond any singular individual experience (Rosenfeld, 2019). This leads to my research question: how do dating apps reflect the political and eugenic implications of preemptive computing and ADM? I argue that preemptive computing and ADM govern visibility and desirability on dating apps, producing political implications by structuring access to potential partners and eugenic implications by reinforcing similarity-based ideals about who should match.

Dating apps mediate romantic relationships in the most fundamental way, by manipulating the agency users have over whom they can meet. Before users choose who to swipe on, platforms choose who users can even see. As Strandburg warns, ADM systems are “inscrutable, making it difficult or impossible to understand how particular decisions are made or to contest them,” thus silently determining who appears in a user’s dating universe (Strandburg, 2019). Dating app algorithms are black boxes; users cannot control or contest who is shown to them, thus denying them the right to see an “unadulterated” dating pool. While the consequences are explored in later paragraphs, the shift in agency is apparent: there is a machine making decisions about who sees you and who you see, therefore preemptively shaping your romantic experience.

Dating app design itself is designed to leech user agency. The curated deck of profiles that users can swipe through is a form of normative architecture. For example, swipe-based interfaces with large *yes* and *no* buttons encourage rapid-fire decisions based on appearance. With hypernudges like decks that adapt to your taste as you scroll and daily “picks” implying algorithmic expertise, dating apps encourage reactive rather than deliberative choice about romantic partners. Fundamentally, the app is designed to be superficial, addictive, and gamified. Using preemptive computing, the platform shifts dating from a double-contingency encounter, where attraction/compatibility develops through interaction, into a single-contingency process by predicting compatibility before a user ever sees another profile. By controlling romantic accessibility within black boxes, dating apps strip users’ intimate agency and lay the groundwork for the systemic biases that follow.

Bias enters dating apps through preemptive computing and the predictive design of ADM, which structures the dating pool before any swiping occurs. Silva explains that “racially

biased outcomes... infect software algorithms and datasets of all types,” because preemptive computing systems/algorithms absorb the bias in their training data (Silva, 2018). These include embedded social theories, which are cultural assumptions about what a good romantic match looks like, such as people with similar education levels or the same race (Silva, 2018). When embedded social theories are treated as data parameters, the app transforms social norms into discriminatory models. Regarding ADM, Levy identifies three pathways through which these systems discriminate, all of which appear in dating apps (Levy, 2017). First, platforms use protected data fields directly, such as race filters (Levy, 2017). Second, they use proxy filters like neighborhood or job titles, which insinuate race and class (Levy, 2017). Finally, design choices like ranking systems can promote dominant norms like thin women or whiteness (Levy, 2017). When dominant norms are promoted by algorithms, they cause unequal romantic accessibility where certain profiles are elevated, while others are pushed down the deck.

Then, Sharabi illustrates how this becomes a self-fulfilling loop. Dating platforms “rely on algorithms and artificial intelligence” to introduce users to partners, and collaborative filtering “can exclude certain groups... by privileging the behaviors of the majority” (Sharabi, 2022). If the majority tends to swipe with a racial preference, the system learns those behaviors as signs of engagement and suppresses minority profiles accordingly. This creates a classification system that ranks users by desirability and represents them according to the patterns the platform has learned to reward. As dating apps convert bias into unequal exposure, they raise questions about the political choices being made around intimacy and romance.

Dating apps have political implications because they make decisions about people: who is shown, who is hidden, who is desirable, and who is not. Strandburg explains that ADM systems can “function as a form of rulemaking,” meaning they have a degree of authority (Strandburg,

2019). On dating apps, this authority determines user visibility on feeds. Dating app visibility determines whether someone is even offered the chance to be considered a romantic partner. As a result, visibility becomes a form of romantic privilege allocated by proprietary algorithms that users cannot see or dispute. Dating apps often justify and strengthen their political power by presenting their matching systems as scientific or data-driven, claiming their models can identify compatibility through match percentages or profile-based assessments (Finkel, 2012). Ironically enough, platforms do not even have the scientific basis they claim. Finkel notes that there is “no compelling evidence” that compatibility models can predict long-term relational success over organic dating (Finkel, 2012). Sharabi shows that dating apps maintain this authority despite contradictory evidence as users experience a placebo effect. They believe that algorithmically assigned matches are meaningful simply because the platform declares them so (Sharabi, 2022). This reinforces the platform’s power to allocate romantic privilege. Ultimately, dating apps make political decisions by essentially controlling access to romantic privilege.

Dating apps convert political implications into eugenic ones as their ranking systems reward certain racialized qualities. Essig notes that some modern dating platforms go beyond surface-level algorithmic curation, offering “genetic and other biometric markers” to determine compatibility, including tests for “symmetry of attraction” (Essig, 2019). Using biology and genetics as a basis for compatibility is eerily reminiscent of language used in eugenic campaigns. When platforms begin treating attraction as something measurable through biological inputs, dating/romance lose their entire premise of human connection. Once intimacy is framed in these pseudo-biological terms, the algorithm is no longer just organizing preferences but actively deciding which bodies count as compatible and which do not. This is a form of soft eugenics, where reproduction between compatible people, or at least deemed so by the model, is not

coerced, but is promoted and encouraged. On the other hand, people/bodies that do not fit “desirable” norms are minimized, thus limiting their reproductive future and freedom.

It is important to note that this soft-eugenic behavior present on dating apps is not due to individual users having romantic preferences. This is a much larger, systemic pattern that is perpetuated by algorithms under the guise of “romantic authority” and compatibility. Williams’ analysis of Match Group’s patented “relevance algorithms” demonstrates that this soft-eugenic logic is seen within mainstream dating apps, as Match Group is the parent company of Tinder, Hinge, and a number of other dating apps. According to the patents, Match Group systems evaluate users using attributes such as “hair color, eye color, height, income, and ethnicity,” transforming physical and socioeconomic traits into algorithmic inputs (Williams, 2024). While these categories are framed as neutral data, they resemble old eugenic systems. Historically, eugenic thinking treated physical traits like height and socioeconomic markers like income as indicators of fitness or genetic quality, used to distinguish between those considered superior and inferior. As these variables are active within the relevance algorithm, they elevate profiles that align with dominant norms of attractiveness, wealth, and racial desirability. This shifts matching from an individualized process into a population-level sorting mechanism, where alignment increases one’s chances of being seen at all. This setup allows people aligned with desirable norms to see and match with each other, while those who are “unaligned” are virtually invisible, thus limiting their romantic privilege and reproductive freedom.

The popular dating platform eHarmony serves as a clear case study of how dating platforms embody soft-eugenic logic. Marketed as a “scientific” dating platform, eHarmony uses proprietary algorithms that users cannot audit. As Essig documents, the platform frames attraction as biologically/genetically grounded and even makes claims about the “probability of

successful pregnancy” when reproducing with matches from their model (Essig, 2019). In doing so, eHarmony presents similarity as a scientifically optimal condition for successful relationships. What is concerning about this narrative is that eHarmony’s branding and success stories are consistently of rich, white, heterosexual couples as the models of relationship success. This reinforces which bodies and lifestyles appear most compatible with the platform’s vision of love, which has strikingly eugenic undertones.

Beyond the political and eugenic implications of online dating, the dating app ecosystem itself is corrupt. Business and user goals are severely conflicted. Dating platforms profit from repeated use, subscription extensions, and ongoing search, while users typically seek fewer matches, stable relationships, and eventual exit from the platform altogether. This creates a structural conflict in which systems are optimized for engagement rather than romantic success, as evidenced by the aforementioned normative architecture and gamified hypernudges. Preemptive computing and ADM support this model by monitoring user behavior and adjusting recommendations to prolong interaction. Sharabi notes that dating technologies “do not simply facilitate interaction but actively structure how users encounter, evaluate, and select potential partners,” emphasizing that algorithmic systems encourage repeated engagement over relationships and “happy endings” (Sharabi, 2022). These systems reward ongoing activity, swiping, messaging, and browsing, because continued engagement produces the data necessary to refine future recommendations. Fundamentally, the user is the product as they produce infinite amounts of data. As Silva argues, digital platforms “are not neutral; the decisions are made by the platform owners for their own benefit,” and algorithmic systems actively “structure social activity” while validating outcomes as objective or “scientifically” chosen (Silva 2018). Users are thus steered toward perpetual searching, not because it serves their romantic interests, but

because prolonged engagement sustains the platform's business model. In this sense, successful matches are not economically preferred for dating apps, revealing that dating apps are not tools for connection, but for-profit infrastructures.

Ultimately, dating apps do more than help people date. Through automated decision-making and preemptive computing, they shape who becomes visible, who is considered desirable, and who is left out of the dating pool entirely. The stakes of dating apps extend beyond individual matches; these systems influence who encounters whom across entire populations, shaping relationship and reproductive patterns and, ultimately, the future of our society. Back at the bar, smiling young men and women approach each other after locking eyes, uninhibited by online "filters" like pre-sorted matches or algorithmic judgments about compatibility. Online, a young man sends a text to his new match. The app says they are 99% compatible. She's the one, he's sure of it.

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